EXPERIMENT REPORT

AffectNet-style expression classification on FER2013

Six training runs and one ensemble on eight FER+ classes

1 Data and protocol

The workshop adapts the AffectNet classification baseline to the original FER2013 images with Microsoft FER+ annotations. The source CSV matches the original Kaggle checksum. Filtering removed images, ties, and examples dominated by unknown or non-face votes yields 26,940 training, 3,396 validation, and 3,376 test examples from the original Training, PublicTest, and PrivateTest partitions. The eight classes are neutral, happy, surprise, sad, angry, disgust, fear, and contempt. FER2013 supplies expression labels only.

All runs use seed 42, AdamW, a OneCycle schedule, inverse-square-root class weighting, data augmentation, and horizontal-flip test-time augmentation. Validation macro-F1 selects each checkpoint. Macro-F1 gives equal weight to each class, including scarce classes such as disgust and contempt. Accuracy measures the fraction of correct predictions. Each configuration was run once.

2 Training configurations

Exercise	Model	Input	Initialization	Epochs	Peak LR
1	ResNet-18 baseline	128×128	ImageNet	25	0.0005
2	Native ResNet-18	48×48	ImageNet	25	0.0005
3	Tuned ResNet-18	128×128	ImageNet	55	0.0003
4a	ResNet-18 from scratch	128×128	Random	25	0.0005
4b	AlexNet	128×128	ImageNet	30	0.0005
4d	Inception-ResNet-v1	160×160	VGGFace2	30	0.0002

Native resolution. A 3×3 stride-1 convolution uses the center of the pretrained 7×7 filters. A 2×2 pool sends a 24×24 map into the first residual stage. The standard stem would reduce a 48×48 input to 12×12 before that stage. The native variant keeps a 3×3 map before global pooling.

Longer schedule. The tuned run uses an exponential moving average (EMA) of the weights with decay 0.995 per batch and learns from FER+ annotator vote shares. Label smoothing drops from 0.10 to 0.02 because the votes already express uncertainty. All runs use the same hard targets for validation and test scoring.

Face recognition pretraining. FaceNet starts from VGGFace2 identity weights, replaces the identity classifier with an eight-class head, and fine-tunes the full network with FaceNet's input standardization. Compared with the baseline, it also changes the backbone, resolution, and learning rate.

3 Test results

Each model was evaluated on the same 3,376 private-test examples. Deltas are in percentage points relative to the baseline.

Exercise	Model	Best epoch	Val macro-F1	Test macro-F1	Test accuracy	Δ macro-F1	Δ accuracy
1	ResNet-18 baseline	21	76.37%	72.32%	85.49%	0.00	0.00
2	Native 48×48	13	75.05%	70.89%	83.23%	−1.43	−2.25
3	Longer schedule and EMA	16	75.32%	69.50%	84.57%	−2.82	−0.92
3b	Equal-weight ensemble	–	**77.24%**	73.29%	**87.23%**	+0.97	**+1.75**
4a	ResNet-18 from scratch	23	72.76%	67.78%	82.49%	−4.54	−2.99
4b	AlexNet	26	69.28%	68.24%	82.32%	−4.08	−3.17
4d	VGGFace2 FaceNet	25	76.50%	**73.99%**	86.49%	**+1.67**	+1.01

Resolution and class imbalance

Baseline F1 was 94.9% for happy, 45.0% for disgust, and 48.0% for contempt. The test set has only 22 disgust and 26 contempt examples, so a few predictions can noticeably change macro-F1. The native run improved disgust F1 to 47.8% and contempt F1 to 51.3%, while fear F1 dropped from 62.1% to 57.0%. The experiment changes the input and stem together. Upsampling to 128×128 gives pretrained filters a denser grid while interpolating the existing pixels.

Accuracy improvement

For the 55-epoch recipe, validation macro-F1 peaked at epoch 16 while training loss continued to fall. Disgust recall rose from 40.9% to 59.1%, but precision fell from 50.0% to 25.0%. The changes in targets and smoothing affect the scale of the reported loss.

The second attempt averages the class probabilities from the baseline, the native-resolution ResNet-18, and FaceNet. The members use different input processing and pretraining, and averaging can temper confident errors when members disagree. This ensemble reuses existing checkpoints and increases test accuracy by 1.75 pp and macro-F1 by 0.97 pp.

Pretraining

ImageNet initialization improved ResNet-18 test macro-F1 by 4.54 pp and accuracy by 2.99 pp under the same 25-epoch schedule. At epoch 5, validation macro-F1 was 67.7% with ImageNet weights and 48.4% from scratch.

FaceNet achieved the highest test macro-F1. It gained 1.67 pp macro-F1 and 1.01 pp accuracy over the baseline, 6.21 pp and 4.00 pp over random initialization, and 5.75 pp and 4.18 pp over AlexNet. Relative to the baseline, disgust and fear F1 rose to 51.3% and 65.1%, while contempt F1 fell to 46.5%.

4 ResNet-18 versus AlexNet during training

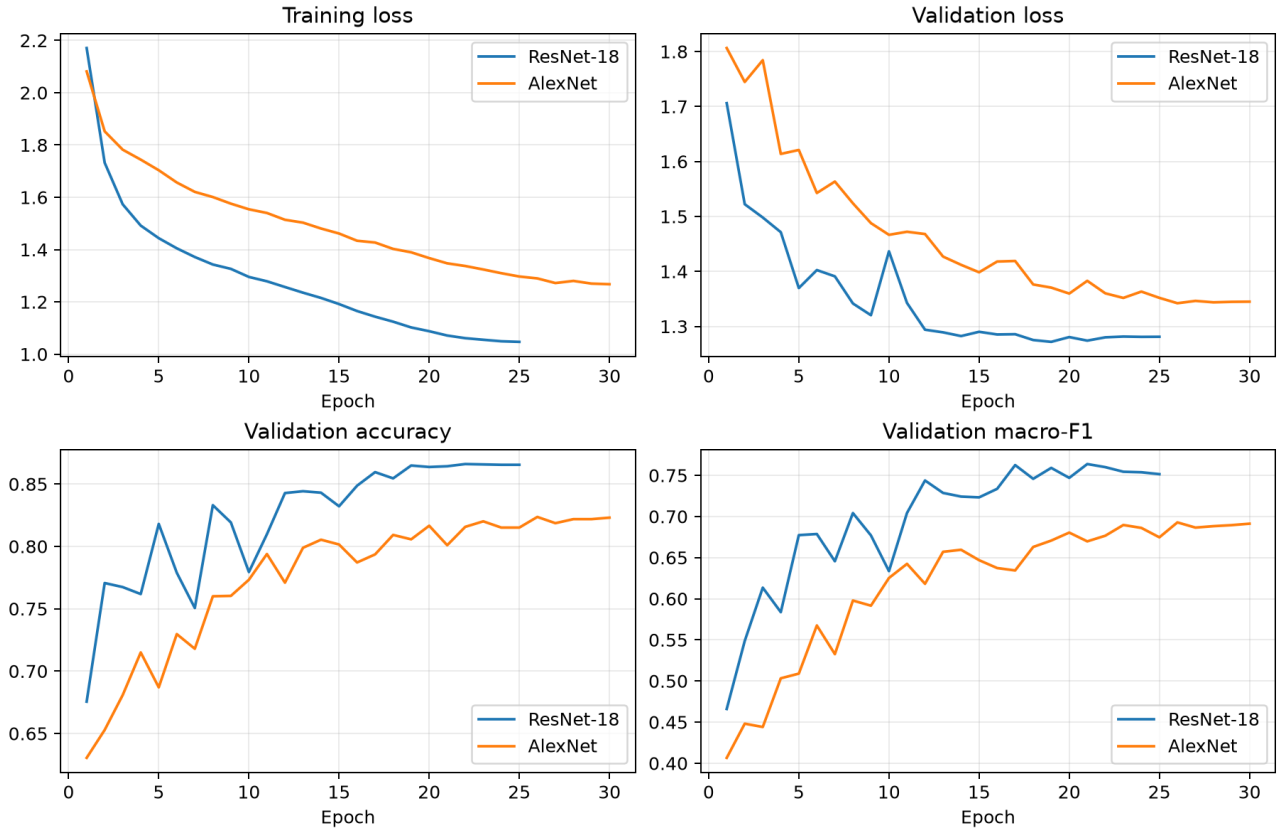


Figure 1. Training loss, validation loss, validation accuracy, and validation macro-F1. Loss is the weighted, smoothed cross-entropy used in both runs.

Epoch	Model	Train loss	Val loss	Val accuracy	Val macro-F1
1	ResNet-18	2.171	1.706	67.6%	46.6%
1	AlexNet	2.081	1.806	63.0%	40.7%
5	ResNet-18	1.444	1.370	81.8%	67.7%
5	AlexNet	1.703	1.621	68.7%	50.9%
25	ResNet-18	1.047	1.281	86.5%	75.1%
25	AlexNet	1.297	1.352	81.5%	67.5%

ResNet-18 reduced the loss more quickly and sustained higher validation scores. AlexNet peaked at 69.28% validation macro-F1 at epoch 26, compared with 76.37% at epoch 21 for ResNet-18. Both validation losses levelled off while training losses continued to fall. On test, AlexNet trailed by 4.08 pp macro-F1 and 3.17 pp accuracy.

ResNet-18 uses residual connections and batch normalization, which can help fine-tune deeper features. AlexNet starts with an 11×11 stride-4 convolution and downsamples early, which can discard subtle local features. Its 57.0 million parameters are concentrated in the fully connected classifier, versus 11.2 million for ResNet-18. The runs also differ in their 25- and 30-epoch OneCycle schedules.

5 Published accuracy references

Label protocol	Study and evaluation note	Reported score
FER2013, 7 classes	IET 12-layer CNN (2025), abstract maximum and happy-class score	99.20%
FER2013, 7 classes	ResNet50–Transformer (2026), stated overall score	89.70%
FER2013, 7 classes	Global–local ensemble (2026), 7,178-image folder test split	76.51%
FER2013, 7 classes	SFER-MDFAE (2024)	76.18%
FER2013, 7 classes	VGG-based single model (2021), no extra training images	73.28%
FER+	PAtt-Lite comparison (2025), abstract score	96.20%
FER+, 8 classes	FER-EMFormer (2026)	95.78%

The highest FER2013 headline number in the reviewed papers, 99.20%, is also the happy-class score in Section 4.1. The paper separately reports 98.20% as a class mean and applies a confidence filter in its detailed evaluation. The paper claiming 89.70% overall describes at least 991 errors across seven confusion-matrix cells. That exceeds the roughly 370 errors compatible with 89.70% on the 3,589-image official private partition.

The PAtt-Lite abstract also reports 68.40% balanced accuracy, and its full protocol was unavailable in the public preview. FER-EMFormer uses 224×224 inputs and MS-Celeb-1M pretraining. The workshop ensemble's 87.23% can be placed alongside those FER+ results with differences in filtering and pretraining in mind. Seven-class FER2013 uses different labels and sample eligibility.

6 Photo inference

The ensemble had the highest validation macro-F1 (77.24%) and was selected for photo inference. OpenCV detected and cropped a face in each of the three photos. The top expression scores were neutral (73.0%), happy (72.9%), and neutral (75.5%). The public report records the scores. The original photos and visual predictions are kept local.

7 Reproducibility

The original runs used Python 3.12.16, PyTorch 2.11.0+cu128, torchvision 0.26.0+cu128, facenet-pytorch 2.6.0, and OpenCV 4.14.0 on an RTX 4090. The workshop repository includes training and inference code, per-epoch histories, aggregate plots, and the test metrics for each configuration.